PYTHON WINTER SPRINT 2026

CODING COUNCIL
a student led tech council

W3B JMI

Core Problem of 9th Jan 2026 : RPS Engine – Rock Paper Scissors

Each Day have one constant theme and three different tracks : Foundation Track especially due to W3B Collaboration, Core Track for those following since DAY-1 and the Project Track for those who have their basics of Python Cleared, they will make these end-to-end projects

FOUNDATION TRACK (FOR ABSOLUTE BEGINNERS AND NEW JOINERS)

[Expected Time to Finish – 1.5hrs]

Resource: https://www.youtube.com/watch?v=UrsmFxElp5k from 3hrs 09 minutes till 3hrs 58 minutes

Core Topic: Conditional Expressions

🎯 Objective

Build a simple Rock–Paper–Scissors game where:

- The user chooses a move
- The computer move is predefined
- The winner is decided using if / elif / else

This task focuses on decision-making using conditionals.

🧠 Problem Statement

Write a Python program that:

1. Asks the user to enter their move:
   - rock
   - paper
   - scissors
2. Use a predefined computer move: 'computer_move = "rock"'
   [Optionally generate random move]
3. Compare the user's move with the computer's move and print:
   - "You Win"
   - "You Lose"
   - "It's a Draw"
4. Print both moves before announcing the result.

Sample Input:

```
Enter your move (rock/paper/scissors): paper
```

Sample Output:

```
Your Move      : paper
Computer Move  : rock
Result         : You Win
```

Beginners Please ask in Group if you get to encounter any problem, any kind of problem in this, or just DM me Every One was once a beginner !!

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CORE TRACK (FOR STUDENTS WHO HAVE BEEN LEARNING FROM DAY 1)

Core Topic: Object Oriented Programming

Resource: https://www.youtube.com/watch?v=UrsmfxEIp5k from 6hr 42 minutes till 7hr 23 minutes

[Expected Time to Finish - 1.5hrs]

🧩 What You Need to Do

Create a Class

Create a class called RPSGame to represent the game.

Inside the class, store:

- Player name
- Player score
- Computer score

Initialize the Game

Use __init__() to:

- Save the player name
- Set both scores to 0

Play One Round

Create a method play_round() that:

- Takes the player's move
- Generates the computer's move
- Uses if / elif / else to decide the winner
- Updates scores

Show Scores

Create a method show_score() to print final scores.

Run the Game

Outside the class:

- Ask for player name
- Ask how many rounds to play
- Use a loop to call play_round() that many times
- Call show_score() at the end

Sample Run:

```
Enter player name: Arshad
Enter number of rounds: 3

Enter your move (rock/paper/scissors): rock
Computer chose: scissors
Result: You Win

Enter your move (rock/paper/scissors): paper
Computer chose: paper
Result: Draw

Enter your move (rock/paper/scissors): scissors
Computer chose: rock
Result: You Lose
```

```
--- SCOREBOARD ---
Arshad : 1
Computer : 1
```

PROJECT TRACK (FOR ADVANCED PARTICIPANTS, MINI PROJECT)

[Expected time to Finish – 3-4 hours]

🎯 Core Objective

Build a CLI-based RPS Engine that includes:

- User authentication (login/register)
- Persistent player profiles
- Match-based gameplay
- Rule-based decision engine
- File-backed storage

🔐 Authentication System (Mandatory)

Features:

- User registers with:
  - username
  - password
- User can log in
- Credentials stored in a file (users.json or users.txt as you wish)

Rules:

- Username must be unique
- Incorrect login → access denied

👤 Player Profile Management

Once logged in:

- Load player profile from file
- Track:
  - Matches played
  - Wins
  - Losses
  - Draws
- Update stats after every match
- Save back to file

File example:

{"arshad": {"password": "1234","wins": 3,"losses": 2,"draws": 1}}

🎮 Game Engine (Object-Oriented Design)

The game must be designed using Object-Oriented Programming (OOP) so that each responsibility is handled by a separate class. This makes the system modular, readable, and easy to extend.

🧑‍💻 User Class

This class represents a registered player.

Responsibilities:

a. Store user information: Username, Password, Wins, losses, draws
b. Load user data from file after login
c. Update statistics after each match
d. Save updated data back to file

👉 Think of this class as the player profile manager.

PROJECT TRACK (FOR ADVANCED, SYSTEM-LEVEL PROBLEM SOLVING)

🎮 RPSEngine Class

This class contains the core game logic.

Responsibilities:

Define valid moves (rock, paper, scissors)

Generate the computer's move

Apply game rules to decide:

Win, Loss, Draw

Return round results to the session controller

👉 This class is the rule engine — it decides outcomes, not user flow.

🔁 GameSession Class

This class manages one complete match.

Responsibilities:

Ask how many rounds to play

Control the round loop

Call the RPSEngine for each round

Track: Round results, Match-level score

Decide final match result & Notify the User class to update statistics

- 👉 This class acts as the orchestrator between user and engine.

🔁 Match System

- Ask number of rounds
- Play round-by-round
- Track live score
- Decide final match result
- Save stats automatically

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PROJECT TRACK – SAMPLE RUN

Sample Run 1: New User Registration + Match

yaml

```
=== RPS ENGINE ===
1. Login
2. Register
Choose option: 2

Create username: arshad
Create password: 1234
Registration successful.

=== LOGIN ===
Username: arshad
Password: 1234
Login successful.

Enter number of rounds: 3

Round 1
Your move (rock/paper/scissors): rock
Computer move: scissors
Result: Win

Round 2
Your move (rock/paper/scissors): paper
Computer move: paper
Result: Draw

Round 3
Your move (rock/paper/scissors): scissors
Computer move: rock
Result: Lose

--- MATCH SUMMARY ---
Player        : arshad
Wins          : 1
Losses        : 1
Draws         : 1
Match Result  : Draw

Profile saved successfully.
```

Sample Run 2: Existing User Login + Match

yaml

```
=== RPS ENGINE ===
1. Login
2. Register
Choose option: 1

Username: arshad
Password: 1234
Login successful.

Enter number of rounds: 2

Round 1
Your move (rock/paper/scissors): paper
Computer move: rock
Result: Win

Round 2
Your move (rock/paper/scissors): rock
Computer move: scissors
Result: Win

--- MATCH SUMMARY ---
Player        : arshad
Wins          : 2
Losses        : 0
Draws         : 0
Match Result  : Win

Profile updated successfully.
```

Sample Run 4: Invalid Move Handling

powershell

```
Enter your move (rock/paper/scissors): fire
❌ Invalid move. Please choose rock, paper, or scissors.
```